SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

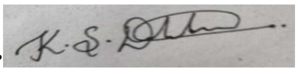
Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I K.S.Dharshana(192412015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

Solution for above answers:

1. Backtracking Search – Doctor Shift Assignment

Problem Statement
Assign 3 doctors (**D1, D2, D3**) to 3 shifts (**Morning, Afternoon, Night**) while satisfying all constraints.

Given Constraints

- **C1:** D1 cannot work Night.
- **C2:** D2 must work before D3.
- **C3:** Only one doctor per shift.
- **C4:** D3 cannot work Morning.
- **C5:** Every doctor must be assigned exactly one shift.

Step 1: Variables
Doctors: D1 ,D2 ,D3
Available Shifts: Morning (M) ,Afternoon (A) ,Night (N)

Step 2: Initial Assignment
Initially, no doctor is assigned.

Step 3: Start Backtracking
Assign D1
Possible shifts: Morning ,Afternoon
Choose **Morning**.
Current Assignment:

Doctor	Shift
D1	Morning
D2	-
D3	-

Constraint Check:D1 not Night ,No duplicate shift
Proceed.

Assign D2
Available shifts: Afternoon ,Night
Try **Night** first.
Current Assignment:

Doctor	**Shift**
D1	Morning
D2	Night
D3	-

Assign D3
Remaining shift:Afternoon
Check constraints:D3 cannot be Morning ,D2 before D3
D2 = Night
D3 = Afternoon
Night is **after** Afternoon.
Constraint C2 violated.
Invalid assignment.
Backtrack.

Backtrack to D2
Instead of Night, assign **Afternoon**.
Current Assignment:

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	-

Assign D3
Remaining shift: Night
Constraint Check D3 not Morning ,D2 before D3
Afternoon < Night
Only one doctor per shift
Every doctor assigned

Final Assignment

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

Backtracking Search Tree

Start

↓

D1 = Morning

↓

D2 = Night

↓

D3 = Afternoon(fail)

↓

Backtrack

↓

D2 = Afternoon

↓

D3 = Night (goal state)

Final Valid Schedule

Shift	**Doctor**
Morning	D1
Afternoon	D2
Night	D3

Hence, the valid schedule obtained using Backtracking Search is:
Morning → D1
Afternoon → D2
Night → D3

2. Robot Path Finding using BFS Search

Since the question mentions a **5×5 grid** but does not specify obstacle positions, assume the following standard grid.

S	0	0	X	0
0	X	0	X	0
0	0	0	0	0
X	X	0	X	0
0	0	0	0	G

Legend

- S = Start (0,0)
- G = Goal (4,4)
- X = Obstacle
- 0= Free Cell

Manhattan Distance
Formula:$h(n) = | x_g - x | + | y_g - y |$

Goal = (4,4)
Example
For Start (0,0)

$$\boxed{h = | 4 - 0 | + | 4 - 0 | = 8}$$

BFS Algorithm
Step 1
Queue
[(0,0)]
Visited
(0,0)
Expand
(0,0)
Possible moves

- Right → (0,1)
- Down → (1,0)

Queue
[(0,1),(1,0)]

Step 2
Expand
(0,1)
New node
(0,2)
Queue
[(1,0),(0,2)]

Step 3
Expand
(1,0)
New node
(2,0)
Queue
[(0,2),(2,0)]

Step 4
Expand
(0,2)
New node
(1,2)
Queue
[(2,0),(1,2)]

Step 5
Expand
(2,0)
New node
(2,1)
Queue
[(1,2),(2,1)]

Step 6
Expand
(1,2)
New node
(2,2)
Queue
[(2,1),(2,2)]

Step 7
Expand
(2,1)
No better node.
Queue
[(2,2)]

Step 8
Expand
(2,2)
New nodes
(2,3)
(3,2)
Queue
[(2,3),(3,2)]

Step 9
Expand
(2,3)
New node
(2,4)
Queue
[(3,2),(2,4)]

Step 10
Expand
(3,2)
New node
(4,2)
Queue
[(2,4),(4,2)]

Step 11
Expand
(2,4)
New node
(3,4)
Queue
[(4,2),(3,4)]

Step 12
Expand
(4,2)
New node
(4,3)
Queue
[(3,4),(4,3)]

Step 13
Expand
(3,4)
Goal found
(4,4)

Priority Queue (Exploration Queue) Updates

Step	**Queue**
1	[(0,1),(1,0)]
2	[(1,0),(0,2)]
3	[(0,2),(2,0)]
4	[(2,0),(1,2)]
5	[(1,2),(2,1)]
6	[(2,1),(2,2)]
7	[(2,2)]
8	[(2,3),(3,2)]
9	[(3,2),(2,4)]
10	[(2,4),(4,2)]
11	[(4,2),(3,4)]
12	[(3,4),(4,3)]
13	Goal

Optimal Path
S
↓

(1,0)
↓

(2,0)
→

(2,1)
→

(2,2)
↓

(3,2)
↓

(4,2)
→

(4,3)
→

Goal

Total Cost
Each move = 1
Total moves = **8**
Therefore,
Optimal Cost = 8

3. Autonomous Rescue Robot

(a) Analysis of Constraints and Their Effect on Decision Making
The rescue robot operates in a disaster-affected building where conditions change frequently. Every constraint directly affects how the robot plans its path.
1. Movement Constraint
The robot can only move in four directions:

- Up
- Down
- Left
- Right

Effect:
The robot cannot move diagonally, so it must carefully choose among the available neighboring cells while planning its route.

2. Obstacles
Some rooms are blocked due to debris or collapsed walls.
Effect:
The robot must detect these blocked cells and avoid them completely. If an intended path is blocked, it has to search for an alternative route.

3. Limited Sensing Capability
The robot can observe only adjacent cells.

Effect:
The robot does not know the entire environment initially. As it moves, it continuously discovers new information and updates its internal map. This makes the problem an **online search** problem rather than an offline one.

4. Movement Cost
Each movement costs **1**.

Effect:
The robot should avoid unnecessary wandering and aim for shorter paths to reduce travel time and energy consumption.

5. Risky Zones
Certain rooms are dangerous and add an extra cost of **2**.

Effect:
Although risky zones are passable, they increase the overall travel cost. The robot prefers safer routes unless passing through a risky area significantly shortens the path.
For example:
Safe cell:
Cost = 1
Risky cell:
Cost = 1 + 2 = 3

6. Dynamic Knowledge Update
The robot gains new information only while moving.

Effect:
The robot repeatedly updates its map, revises its path if new obstacles are detected, and replans when necessary. This allows it to adapt to changing environments.

(b) Uniform Cost Search always expands the node with the **lowest cumulative path cost**. Since the environment contains risky cells with higher costs, UCS naturally finds the minimum-cost route instead of just the shortest route.
Step 1: Initialize
- Place the start node **S** in the priority queue with cost 0.
- Create an empty explored list.

Step 2: Expand the Lowest-Cost Node
- Remove the node with the smallest accumulated cost.
- Check whether it is the goal.

Step 3: Sense the Environment
- Observe adjacent cells.
- Mark blocked cells as obstacles.

- Mark risky cells with their additional cost.

Step 4: Generate Successors
For each valid neighboring cell:
- Ignore blocked cells.
- Calculate the new cumulative cost:
  - Safe cell: Current Cost + 1
  - Risky cell: Current Cost + 3
- Insert the new node into the priority queue if it offers a lower-cost path.

Step 5: Repeat
Continue selecting the lowest-cost node, sensing the surroundings, and updating the map until the goal is reached.

Step 6: Goal Reached
When the goal node is removed from the priority queue, the robot reconstructs the least-cost path and rescues the survivor.

(c) Application of AI Concepts

Intelligent Agent
The robot is an intelligent agent because it:
- Perceives the environment through sensors.
- Maintains an internal map.
- Chooses actions based on current knowledge.
- Acts autonomously without constant human control.

Rationality
A rational agent selects actions that maximize its performance. In this scenario, the robot:
- Avoids obstacles.
- Prefers safe routes.
- Minimizes the total travel cost.
- Replans when new information becomes available.

Thus, every action is chosen to maximize the chances of a successful rescue while minimizing risk and cost.

Environment Type
The rescue scenario can be classified as:
- **Partially Observable:** Only adjacent cells are visible.
- **Dynamic:** The environment may change as new obstacles are discovered.
- **Sequential:** Each action influences future decisions.
- **Stochastic (or Uncertain):** Outcomes may vary due to incomplete information.
- **Single-Agent:** Only one rescue robot is making decisions.
- **Discrete:** The environment consists of separate grid cells and discrete actions.

(d) Justification for the Chosen Approach
Uniform Cost Search combined with an Online Search Agent is well suited because it minimizes the total path cost, correctly handles different movement costs (safe vs. risky zones), and adapts as new information becomes available. Unlike BFS, which only minimizes the number of steps, UCS accounts for varying costs and therefore selects safer routes when they reduce the overall cost. The online search capability enables the robot to continuously update its knowledge of the environment, avoid newly discovered obstacles, and replan efficiently. This combination improves safety, conserves energy, and increases the likelihood of successfully reaching the survivor in an uncertain disaster environment.

Conclusion

By combining **Uniform Cost Search** with an **Online Search Agent**, the rescue robot can navigate a partially observable and dynamic environment efficiently. It intelligently senses nearby cells, updates its knowledge, avoids obstacles and risky areas when beneficial, and always aims to minimize the total rescue cost. This approach ensures safe, rational, and adaptive navigation, making it highly suitable for real-world disaster response scenarios.